

# MICHELE DE CILLIS

## Front-End Engineer

✉ work@demic.dev   📍 Milan, IT   🔄 demic-dev   🔗 demic-dev   🌐 demic.dev

## EXPERIENCE

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### Research Intern

#### NERDS Lab @ ITU

📅 Sep 2025 – Dec 2025

📍 Copenhagen, DK

Conducted bachelor's thesis research on measuring polarization in large-scale directed social networks during an Erasmus traineeship. Stack: Python, NumPy, PyTorch, Pandas.

- Measured polarization on directed networks by applying the Magnetic Laplacian — the **first use in this context**
- Earned the **highest thesis grade (7/7)**. Read [here](#)

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### Co-founder

#### Polyneta

📅 Apr 2025 – Dec 2025

📍 Remote

Co-founded a startup connecting companies that sell to municipalities to simplify direct-award procurement, running customer discovery and building the early product with AI-accelerated development.

- Built the alpha (pre-MVP) aggregating company services for municipalities, using AI to accelerate development
- Ran customer discovery through **3-5 exploratory interviews** and Mom Test surveys

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#### Pane&Design (PED, part of humanvsmachine.co)

📅 Oct 2021 – Dec 2022

📍 Milan, IT

Built and optimized high-traffic e-commerce sites and internal telemedicine tools for a Fortune 500 client at a digital consulting agency, using React, TypeScript, styled-components, Tailwind, and Material UI.

- Improved client-side performance, cutting re-renders and load times by **perceived 50%** on a high-traffic e-commerce site ( **100k monthly visits**), via microservices with dynamic JS loading
- Built a telemedicine tool for the client's ophthalmologists, unifying a multi-section medical history form into a single-page flow with react-jsonschema-form
- Contributed to further high-traffic client sites (**up to 200k monthly visits**) and to the agency's own website

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### Full-Stack Engineer

#### Online Impresa (part of areaweb.org)

📅 Jul 2020 – Sep 2021

📍 Bari, IT

Sole developer at a small software agency serving SMEs, building the company's flagship mobile app platform from scratch with TypeScript, React Native, GraphQL, and Expo, and working directly with clients on requirements.

- Built the company's flagship cross-platform mobile app from scratch as sole developer with React Native, TypeScript, GraphQL, and Expo; shipped to the Play Store and App Store
- Delivered **100 client apps**, gathering and implementing requirements directly with clients
- Built internal tools, including a WhatsApp bot posting every closed order to a live office dashboard

## PROJECTS

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### Open Source Contributions

📅 2025 – Present

- Usability bugfixes (mainly state sync) in [Pixelfed](#) (**+7k stars**, React Native)
  - Improved the Italian translation in [Ghost Blog](#) (**+54k stars**)
  - Ported and maintaining a package in [nixpkgs](#) (**+25k stars**, Nix language)
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### Personal Homelab

Self-host personal services (Nextcloud, Immich, and more) on a VPS with NixOS, using reproducible configuration-as-code to add services and their infrastructure (backups, nginx) quickly, with the same config shared across devices. Uses Tailscale for VPN, nginx as a reverse proxy, and daily BorgBase backups.

📅 Mar 2025 – Present

### ALS Biomarker Identification

Analyzed RNA-Seq data and developed a model that generalizes with 90% accuracy for the ALS group. Authored comprehensive documentation of a workflow to identify potential biomarkers.

📅 May 2024 – May 2024

## EDUCATION

### BSc Computer Science Università degli Studi di Milano

📅 Sep 2022 – Apr 2026

Calculus   Linear Algebra   Physics   Data Structures and Algorithms

### Erasmus+ Exchange Program Université Paris-Saclay

📅 Sep 2023 – Jun 2024

Bio Informatics   Machine Learning & NLP   Linear Programming

## SKILLS

Language and Tools - TypeScript   JavaScript   HTML/CSS   GraphQL   Figma   Git   Python   Nix   Bash   Linux   Docker   Typst  
LaTeX

Frameworks and Libraries - React   React Native   styled-components   Tailwind   Material UI   TanStack Query   Node.js   NumPy  
DuckDB   PyTorch   Matplotlib   scikit-learn

## ACCOMPLISHMENTS

### Top-Percentile Academic Ranking University of Milan

📅 2024/25

Ranked in the top 20% for academic performance and the top 15% for graduation grade within the reference academic year.

### First Place University Startup Challenge

📅 2025

Won first place as a team of 4 by conducting market research, gathering early feedback, and building the pitch, landing page, and Figma prototype for what became Polyneta.

### Silicon Valley Study Tour

📅 Aug 2025

Selected for a competitive program visiting leading companies and universities (Cisco, Apple, Google, LinkedIn, Electronic Arts, PureStorage, Sysdig, Zoox, Stanford, Berkeley), attending keynotes from their teams and building connections.

## CERTIFICATIONS

Cambridge C1 Advanced · 📅 26 Mar 2026

# LANGUAGES

|         |                |
|---------|----------------|
| Italian | Native speaker |
| English | C1 Advanced    |
| French  | B1             |
| Spanish | B1             |